The Cohesion–Integration Problem

A Mathematical Theory of
Central Bank Independence, Coordination and Merger

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Abstract

This paper develops a compact framework for choosing among independent central banks, coordinated but separate central banks, and a merged central bank. It combines welfare economics with international-relations (IR) variables, endogenous cohesion, strategic commitment, financial transmission, and institutional adaptation. A quadratic benchmark makes the three regimes analytically comparable; a dynamic extension permits cohesion and institutional mismatch to evolve. A reproducible simulation design is specified, while illustrative numerical values are explicitly distinguished from empirical evidence. The framework offers conditional propositions and testable hypotheses, not a claim that monetary merger is universally optimal. Its central implication is that institutional integration should be selected by comparing total benefits, distributional effects, strategic reliability, and adjustment costs rather than treating cohesion as a sufficient statistic.

The paper ends with "The End"

Keywords: central bank independence; monetary coordination; monetary union; international relations; cohesion; international finance; welfare economics; institutional adaptation.

1 Introduction

Monetary institutions are simultaneously economic mechanisms and political institutions. Their design affects stabilization, commitment, distribution, and the allocation of authority [1, 5, 6]. This paper builds directly on Ghosh's transaction-oriented framework for central-bank merger and acquisition, which organizes monetary integration around valuation, synergies, balance-sheet harmonization, risk assessment, and implementation; it also draws on his analysis of dual government to treat integration as a problem of distributing and constraining authority [10, 13]. The present paper differs by placing independence, coordination, and merger in a common welfare comparison and by making international-relations conditions and cohesion explicit. Optimum-currency-area theory emphasizes adjustment to asymmetric shocks, while later work stresses that trade and business-cycle correlations may themselves change after integration [1, 2]. Political-economy accounts likewise treat monetary arrangements as choices shaped by sovereignty and exposure to external shocks [3, 6].

The question is not whether integration is intrinsically good. It is: *under what conditions should nations retain independent central banks, coordinate policy while retaining separate institutions, or merge their central banks?* The answer must include international relations because common monetary institutions depend on bargaining, strategic alignment, credible commitments, and the political legitimacy of shared decisions. Economic interdependence alone does not establish these conditions.

2 Institutional alternatives and cohesion

Let $j \in \{I, C, M\}$ denote independent central banks, coordinated but separate central banks, and a merged central bank. Let $c_t \in [0, 1]$ be a composite cohesion index. One candidate specification is

$$c_t = \omega_E E_t + \omega_P P_t + \omega_S S_t + \omega_B B_t, \qquad \omega_k \geq 0, \quad \sum_k \omega_k = 1, \tag{1}$$

where E_t measures economic and financial compatibility, P_t political compatibility, S_t strategic or security alignment, and B_t the credibility of bargaining and commitments. Components should be normalized

before aggregation. Alternative aggregation rules and minimum constraints should be tested: a high score in one dimension need not compensate for a critical failure in another.

For N nations, pairwise cohesion may be summarized by

$$c_t = \frac{2}{N(N-1)} \sum_{i<j} r_{ij,t}, \tag{2}$$

where $r_{ij,t} \in [0,1]$. This mean is not a sufficient statistic: identical averages can conceal different alliance structures, veto players, and vulnerable minorities.

Table 1: Institutional alternatives and principal trade-offs

Arrangement	Potential benefit	Principal cost or risk
Independent (I)	National tailoring, local accountability, insulation from partners' shocks.	Duplicated capacity, policy spillovers, exchange-rate and coordination frictions.
Coordination (C)	Internalizes selected spillovers while retaining separate institutions and exit options.	Bargaining delay, free-riding, weak enforcement, coordination failure.
Merger (M)	Common stance, fewer intra-union monetary frictions, pooled credibility where commitments are trusted.	Loss of national instruments, asymmetric-shock exposure, distributional conflict, transition costs.

3 A minimal welfare model

Let social welfare under arrangement j be

$$W_j(c,\theta) = G_j(c,\theta) - L_j(c,\theta) - T_j, \tag{3}$$

where G_j denotes stabilization and coordination gains, L_j expected losses from heterogeneity, weak commitment, distributional conflict, and institutional risk, T_j transition and governance costs, and θ other conditions such as shock asymmetry, fiscal risk-sharing capacity, and credibility. Welfare is social, not merely the central bank's objective; transfers must not be confused with aggregate resource gains. The optimal feasible arrangement is

$$j^*(c,\theta) \in \operatorname*{argmax}_{j \in \{I,C,M\}} W_j(c,\theta). \tag{4}$$

For a transparent benchmark, suppose incremental net welfare relative to independence is

$$D_C(c) = W_C(c) - W_I(c) = a_C c - b_C, \tag{5}$$

$$D_M(c) = W_M(c) - W_I(c) = a_M c - b_M - d_M(1-c)^2, \tag{6}$$

where all parameters are nonnegative. Coordination gains rise with cohesion in this benchmark, while merger incurs a convex penalty when cohesion is low. These are modelling assumptions, not universal laws.

Proposition 1 (Conditional coordination threshold)**.** *If $a_C > 0$, coordination weakly dominates independence exactly when $c \geq c_C = b_C/a_C$, provided $c_C \in [0,1]$.*

Proof. By equation (5), $W_C(c) - W_I(c) \geq 0$ if and only if $a_C c - b_C \geq 0$. Since $a_C > 0$, this is equivalent to $c \geq b_C/a_C$. If the ratio lies outside $[0,1]$, the threshold is outside the feasible domain. □

The merger threshold is determined by the roots, if any, of $D_M(c) = 0$. Because this is quadratic in c, there may be zero, one, or two real roots; feasibility and dominance over coordination must still be checked. The benchmark therefore does not guarantee a unique independence–coordination–merger sequence.

Proposition 2 (A genuine intermediate coordination regime)**.** *Suppose $W_C(c) > W_I(c)$ and $W_C(c) > W_M(c)$ throughout a non-empty interval $J \subseteq [0,1]$. Then coordination is optimal on J. If independence dominates on a lower-cohesion interval and merger dominates on a higher-cohesion interval, the model admits three distinct optimal regimes.*

Proof. The result follows from the pairwise inequalities defining the maximizer in equation (4). The assumptions are sufficient, not necessary. □

4 International relations: strategic credibility and distribution

The IR extension treats nations as strategic actors rather than passive economic regions. Let country i's payoff from arrangement j be

$$U_{ij,t} = u_{ij,t}^{\text{econ}} + u_{ij,t}^{\text{security}} + u_{ij,t}^{\text{bargaining}} - u_{ij,t}^{\text{sovereignty}}. \tag{7}$$

A collective arrangement is politically sustainable only if participation constraints hold, for example

$$U_{ij,t} \geq U_{iI,t} - \kappa_i, \tag{8}$$

where $\kappa_i \geq 0$ represents tolerated short-run losses or agreed compensation. A fuller model would endogenize voting weights, emergency powers, transfers, and exit provisions. The collective welfare maximum is not implementable if pivotal members reject it.

The variable F_t is informed by Ghosh and ChatGPT's related framework, where institutional "fakeness" is a technical measure of the gap between formal central-bank independence and effective subordination to fiscal, political, strategic, or military objectives, not a synonym for fraud. That manuscript derives different conditional benchmarks for peace and war and treats credibility as institutional capital; here, the concept supplies a state variable for the IR-finance interface rather than a universal prescription [11]. A credibility state $K_t \in [0, 1]$ can evolve as

$$K_{t+1} = \Pi_{[0,1]} \left[(1 - \delta F_t) K_t + \rho(1 - F_t) - \chi Q_t \right], \tag{9}$$

where $F_t \in [0, 1]$ is effective political subordination of the monetary authority, Q_t measures commitment breaches or distributional conflict, and $\Pi_{[0,1]}$ truncates to the unit interval. The reduced form encodes the hypothesis that excessive subordination or repeated breaches may erode credibility; its coefficients require estimation or calibration. The benchmark is consistent with, but does not independently validate, the state-dependent credibility mechanism in Ghosh and ChatGPT's paper on optimal central-bank fakeness [11].

Cohesion is itself endogenous in the candidate law

$$c_{t+1} = \Pi_{[0,1]} \left[(1 - \lambda) c_t + \lambda \bar{c} + \gamma \mathcal{G}_{j_t} - \zeta \mathcal{D}_{j_t} + \varepsilon_{t+1} \right], \tag{10}$$

where $\bar{c}$ is baseline cohesion, $\mathcal{G}_{j_t}$ captures gains from repeated cooperation, $\mathcal{D}_{j_t}$ captures distributional or strategic disputes, and ε_t is a shock. Integration may increase shared interests but can also expose conflict over adjustment costs [2, 4].

5 Endogenous adaptation and institutional purpose

Let x_t represent an institution's operating configuration and e_t its environment. Define mismatch $m_t = x_t - e_t$. Under a stationary environment, the adaptation law

$$x_{t+1} = x_t + \eta(e_t - x_t) \tag{11}$$

implies

$$m_{t+1} = (1 - \eta) m_t. \tag{12}$$

Mismatch converges to zero for every initial condition if and only if $0 < \eta < 2$; convergence is monotone for $0 < \eta < 1$, immediate for $\eta = 1$, and oscillatory for $1 < \eta < 2$. This is a scalar stability result, not a claim that all institutional change is beneficial. If the environment drifts by v per period, then $m_{t+1} = (1 - \eta) m_t - v$, and the stable steady-state mismatch is $m^* = -v/\eta$.

Institutional fit must be distinguished from fidelity to institutional purpose. Let q_t denote deviation from the declared mandate. A sustainable configuration may require

$$|m_t| \leq \bar{m} \qquad \text{and} \qquad |q_t| \leq \bar{q}. \tag{13}$$

Survival is not proof of success; stability is not proof of legitimacy; adaptation is not synonymous with improvement. Organizational psychology can inform mechanisms such as learning, imitation, incentives, group identity, and loss aversion, but political institutions should not be reduced to individual psychology.

6 International finance and welfare accounting

Potential transmission channels include cross-border interest-rate differentials, exchange-rate risk, sovereign spreads, capital flows, and contagion. These are predictions, not automatic consequences. A related paper by Ghosh defines Central Bank Performance as the proportional deviation of inflation from target and derives a Central Bank Risk Premium from a stated equilibrium-pricing relation [12]. With $b_t = \pi_t/\pi^* - 1$, the proposed measure is

$$\mathrm{CBRP}_t = \frac{b_t(1+b_t)}{1-b_t}, \qquad \pi^* \neq 0, \quad b_t \neq 1. \tag{14}$$

This construct can motivate a financial outcome or robustness measure in the present framework, but its algebraic derivation from a maintained pricing equation is not, by itself, evidence that it matches observed sovereign bond spreads. That correspondence requires separate empirical validation. A panel specification might be

$$\begin{aligned} y_{it} = \alpha_i + \tau_t + \beta_1 M_{it} + \beta_2 C_{it} + \beta_3 c_{it} + \beta_4 (M_{it} c_{it}) \\ + \Gamma' X_{it} + u_{it}, \end{aligned} \tag{15}$$

where y_{it} could be a sovereign spread or exchange-rate volatility measure, M_{it} and C_{it} identify merger and coordination, and X_{it} contains controls. Endogenous institutional choice, anticipation, selection, and common shocks prevent causal interpretation of ordinary fixed-effects estimates by themselves. Event studies, synthetic controls, instrumental variables, or structural estimation should be used only where identifying assumptions are defensible.

Financial planning should track transition expenditure, fiscal backstops, liquidity facilities, reserve pooling, contingent liabilities, and the distribution of adjustment costs. A schematic welfare identity is

$$W_j = \text{aggregate surplus} - \text{real resource costs} - \text{expected crisis losses} - \text{institutional risk}. \tag{16}$$

Distributional incidence should be reported separately. A transfer can improve participation or legitimacy without constituting a net aggregate gain.

The physics analogy is limited but useful: eigenvalues of a linearized dynamic system can diagnose local stability. It does not follow that nations obey physical laws independent of agency. Biology's adaptation concepts and chemistry's equilibrium metaphors are heuristics, not substitutes for strategic mechanisms. Linguistic framing, public narratives, religious or philosophical legitimacy, and psychological perceptions may affect trust, but should enter empirical work only through defensible operational measures. Geopolitics and warfare economics enter through security shocks, sanctions, emergency financing, strategic dependence, and scarce-resource allocation, not through a presumption that monetary integration is a military instrument.

7 Simulation design

The simulation should be a transparent model experiment, not a disguised empirical result. For each parameter draw, compute welfare, enforce participation constraints, select the feasible maximizer, and update cohesion and credibility.

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for each parameter vector theta:
    initialize cohesion c, credibility K, and institution state x
    for t = 1,...,T:
        compute W_I(c,K,theta), W_C(c,K,theta), W_M(c,K,theta)
        remove arrangements violating participation or feasibility
        choose the feasible arrangement with greatest social welfare
        record regime, welfare, financial outcomes, and distribution
        update credibility K and cohesion c
        update institutional mismatch and purpose deviation
    summarize regime shares, transitions, welfare, and instability
repeat over parameter grids and random seeds
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For a deterministic benchmark, set $W_I(c) = 0$, $W_C(c) = 0.25c - 0.04$, and $W_M(c) = 0.40c - 0.12 - 0.10(1-c)^2$. Evaluating these functions on the displayed grid generates the table below. This is a reproducible numerical illustration of the assumed functions, not a calibrated simulation or empirical estimate.

A robust design should then include one-at-a-time sensitivity analysis, global parameter sampling, alternative cohesion measures, asymmetric shocks, commitment failures, distributional compensation, and switching costs. The transaction and risk-assessment components of Ghosh's merger-and-acquisition framework provide a natural checklist for transition costs and balance-sheet harmonization, while the credibility-stock and regime-switching mechanisms in the optimal-fakeness paper motivate additional peace/conflict sensitivity cases [10, 11]. Code, random seeds, parameter ranges, and outputs should be archived. If confidential payment or supervisory data are used, cybersecurity, access control, provenance, versioning, and adversarial validation must be documented.

Table 2: Computed welfare values from the illustrative deterministic benchmark (not empirical or calibrated output)

Cohesion c	W_I	W_C	W_M
0.10	0.00	-0.015	-0.161
0.30	0.00	0.035	-0.049
0.50	0.00	0.085	0.055
0.70	0.00	0.135	0.151
0.90	0.00	0.185	0.239

These computed values illustrate independence at low cohesion, coordination at intermediate cohesion, and merger at high cohesion under the stated benchmark functions. They do not establish that this ranking holds for any actual group of nations. A final empirical paper would need calibrated or estimated parameters and should publish its implementation.

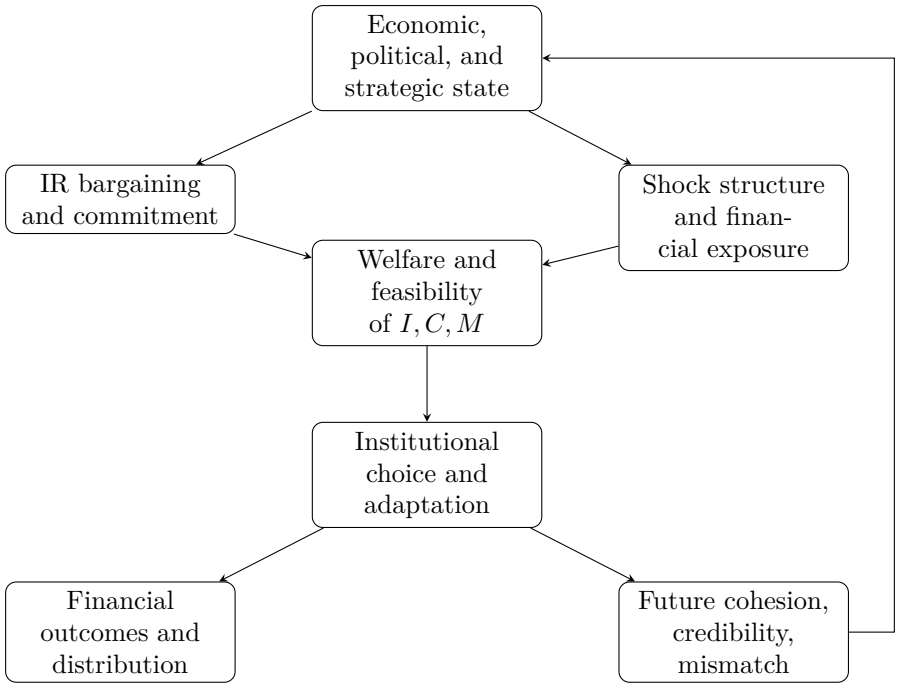


Figure 1: Conceptual architecture linking IR, institutional choice, finance, and adaptation. Arrows denote proposed channels, not established causal effects.

8 Logic, computation, and interdisciplinary scope

The formal argument uses constrained optimization, dynamic systems, and explicit counterexamples. Computationally, begin with deterministic grid search before adding stochastic simulation, agent-based bargaining, or reinforcement learning. Machine learning may forecast state variables or detect nonlinear patterns, but predictive accuracy alone does not identify causal effects. Structural assumptions, out-of-sample validation, calibration uncertainty, and model risk must remain visible.

The relevant disciplines have distinct roles: economics and finance supply welfare and transmission channels; political science, IR, geopolitics, and diplomacy supply bargaining and commitment; statistics and econometrics supply uncertainty and identification; mathematics and physics supply stability analysis; psychology, linguistics, philosophy, and religion may inform trust and legitimacy if operationalized; biology and chemistry provide limited adaptation and equilibrium analogies; computer science supplies reproducible algorithms and secure data handling. Ghosh's treatise on dual government is relevant to the constitutional side of this problem because it examines how authority is distributed horizontally and vertically and how checks, balances, and adaptation affect institutional performance [13]. The analogy is structural: it does not imply that a central-bank merger is itself a system of dual government. Psychiatric diagnoses are neither required nor warranted as explanations of political disagreement. Warfare economics should be represented by observable security shocks, sanctions, emergency spending, and contingent liabilities, with civilian welfare and escalation risks explicit.

9 Limitations and research propositions

The model is parsimonious and its parameters are not estimated here. The numerical table is hypothetical. Cohesion is multidimensional; aggregation can conceal veto players and unequal exposure; welfare weights are normative; and observed institutional arrangements are selected rather than randomly assigned. No causal or country-specific policy conclusion follows from the benchmark alone.

The framework yields testable propositions: (1) conditional on shock structure and governance, higher cohesion may increase the net benefit of coordination or merger; (2) coordination may dominate both independence and merger over a non-empty parameter region; (3) asymmetric shocks and weak fiscal risk-sharing may reduce merger welfare even at high political cohesion; (4) commitment failures can lower credibility and feed back into cohesion; and (5) integration may change the conditions that initially justified it. These are hypotheses to test, not established universal laws [2, 6]. They also specify how this paper extends the author's existing merger, credibility, performance-measurement, and institutional-governance manuscripts rather than treating the current paper as an isolated proposal [10, 11, 12, 13].

10 Conclusion

The central bank integration problem is a choice among institutional arrangements under uncertainty and strategic interdependence. A minimal welfare model clarifies the conditions required for independence, coordination, or merger to dominate. An IR layer adds bargaining and commitment; a dynamic layer lets cohesion, credibility, and mismatch evolve; and the international-finance layer produces testable predictions about spreads, volatility, and capital flows. No institutional form is universally optimal, and cohesion alone is not sufficient to select one. The research programme must derive welfare comparisons, enforce political participation constraints, simulate transparent parameter ranges, and test financial predictions against data. The distinction between proof, simulation, and empirical identification is essential.

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Glossary

Cohesion
A normalized summary of economic, political, strategic, and commitment compatibility.

Coordination
Cooperation among separate central banks without full institutional merger.

Institutional mismatch
Difference between an institution's actual configuration and its environment.

Purpose deviation
Departure from a declared mandate, even when the institution survives.

Participation constraint
A condition requiring a member to prefer joining or remaining to its outside option.

Credibility capital
A state variable summarizing capacity to sustain trusted commitments.

Optimal regime
The feasible institutional arrangement maximizing specified social welfare.

Simulation result
A numerical output conditional on model, parameters, and implementation; not empirical evidence by itself.

Identification
Conditions required to distinguish a causal effect from observationally equivalent explanations.

The End